

Notch filter >>

$$Y_1(n) = b_2 \cdot X(n) + b_1 \cdot X(n-1) + X(n-2)$$

$$Y_2(n) = Y_1(n) + b_3 \cdot Y_1(n-1) + b_4 \cdot Y_1(n-2)$$

$$Y(n) = a_1 \cdot Y(n-4) + a_2 \cdot Y(n-8) + Y_2(n-2) + b_5 \cdot Y_2(n-4) + b_6 \cdot Y_2(n-6)$$

Notch frequency = fn;	a1 = 2*(r^4)*cos(4*nu);
Notch width = r (0<r<1) ;	a2 = -(r^8);
fs = 500M;	b1 = -2*cos(nu);
nu = 2*pi*fn/fs;	b2 = 1;
	b3 = 2*r*cos(nu);
	b4 = r*r;
	b5 = 2*r*r*cos(2*nu);
	b6 = r^4;